



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)
Re-Accredited by NAAC with 'A' Grade

Department of Computer Science and Engineering DevOps Lab - L3

Lab Assignment No: 11	
Name of Student	Aryan Sheladia
PRN	23070122202
Batch	2023 - 2027
Class	CSE – B1
Academic Year & Semester	2026 - 2027, 7th Semester
Title of Assignment	Docker Compose for Multi-Container app
Implementation & Theory	<pre>from flask import Flask import redis app = Flask(__name__) r = redis.Redis(host='redis', port=6379, decode_responses=True) @app.route('/') def hello_world(): count = r.incr('hits') return f'Hello, Docker! This page has been viewed {count} times.' if __name__ == '__main__': app.run(debug=True, host='0.0.0.0', port=5000)</pre> The Flask application from the previous assignment was reused and modified to communicate with a Redis service. The required Python dependencies and Docker configuration were retained, while a Docker Compose file was added to define the multi-container application.

```

1  services:
2      web:
3          build: .
4          ports:
5              - "8080:5000"
6          depends_on:
7              - redis
8
9      redis:
10         image: redis:7-alpine
11         volumes:
12             - redis-data:/data
13
14     volumes:
15         redis-data:

```

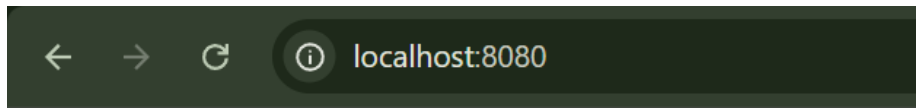
Docker Compose Configuration: Docker Compose defines two services: web for the Flask application and redis for the Redis database. The web service is exposed on port 8080, while Redis is connected to the application through the Compose network. A named volume is used for Redis data persistence.

```

✓ Container assignment-11-redis-1 Created
✓ Container assignment-11-web-1 Created
Attaching to redis-1, web-1
redis-1 | 1:C 21 Aug 2026 09:27:36.297 * o000o000o000o Redis is starting o000o000o000o
redis-1 | 1:C 21 Aug 2026 09:27:36.297 * Redis version=7.4.11, bits=64, commit=00000000,
redis-1 | 1:C 21 Aug 2026 09:27:36.297 # Warning: no config file specified, using the de
redis-1 | 1:M 21 Aug 2026 09:27:36.301 * monotonic clock: POSIX clock_gettime
redis-1 | 1:M 21 Aug 2026 09:27:36.305 * Running mode=standalone, port=6379.
redis-1 | 1:M 21 Aug 2026 09:27:36.306 * Server initialized
redis-1 | 1:M 21 Aug 2026 09:27:36.306 * Ready to accept connections tcp
web-1 | * Serving Flask app 'app' (lazy loading)
web-1 | * Environment: production
web-1 | WARNING: This is a development server. Do not use it in a production deploy
web-1 | Use a production WSGI server instead.
web-1 | * Debug mode: on
web-1 | * Running on all addresses (0.0.0.0)
web-1 | WARNING: This is a development server. Do not use it in a production deploy
web-1 | * Running on http://127.0.0.1:5000
web-1 | * Running on http://172.18.0.3:5000 (Press CTRL+C to quit)
web-1 | * Restarting with stat
web-1 | * Debugger is active!

```

Docker Compose Build and Deployment: Docker Compose builds the Flask web service and starts it together with the Redis service. The two services communicate through the Compose network, forming a multi-container application.



Hello, Docker! This page has been viewed 5 times.

Cross-Service Communication Verified: The Flask web application successfully communicated with the Redis service through the Docker Compose network. Each browser request incremented the Redis hits counter, and the increasing value confirmed successful communication between the two containers.

```
PS E:\DevOps_SEM_7\Assignment - 11> docker compose ps
```

NAME	IMAGE	COMMAND	SERVICE	CREATED	STATUS	PORTS
assignment-11-redis-1	redis:7-alpine	"docker-entrypoint.s..."	redis	8 minutes ago	Up 8 minutes	6379/tcp
assignment-11-web-1	assignment-11-web	"python app.py"	web	8 minutes ago	Up 8 minutes	0.0.0.0:8080->5000/tcp, [::]:8080->5000/tcp

```
PS E:\DevOps_SEM_7\Assignment - 11>
```

Compose Service Status: The docker compose ps command confirms that both the Flask web service and Redis service are running as part of the same multi-container Compose application.

```
PS E:\DevOps_SEM_7\Assignment - 11> docker compose down
```

[+] Running 3/3

- ✓ Container assignment-11-web-1 Removed
- ✓ Container assignment-11-redis-1 Removed
- ✓ Network assignment-11_default Removed

```
PS E:\DevOps_SEM_7\Assignment - 11>
```

Teardown: The Docker Compose application was stopped and the web and Redis containers, along with the Compose-managed network, were removed using docker compose down. The named Redis volume is retained by default, allowing the stored data to persist for future deployments.